Long Title of Thesis

by

XXXXXXXXXX
Registration No.: YYYYYYYYYYY

A thesis submitted to the
Academy of Scientific & Innovative Research
for the award of the degree of
DOCTOR OF PHILOSOPHY
in
SCIENCE

Under the supervision of
Dr. XXXXXXXXX

Name of Institute

Academy of Scientific and Innovative Research
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Ghaziabad, U.P. – 201 002, India

2025

Certificate

This is to certify that the work incorporated in this Ph.D. thesis entitled, **"Long title of Thesis"**, submitted by **XXXXXXXXX** to the Academy of Scientific and Innovative Research (AcSIR) in fulfillment of the requirements for the award of the Degree of **Doctor of Philosophy**, embodies original research work carried-out by the student. We, further certify that this work has not been submitted to any other University or Institution in part or in full for the award of any degree or diploma. Research materials obtained from other sources and used in this research work have been duly acknowledged in the thesis. Images, illustrations, figures, tables, etc., used in the thesis from other sources, have also been duly cited and acknowledged.

(Signature of Student)
Name: **XXXXXXXX**

Date:

(Signature of Supervisor)
Name: **Dr. XXXXXXXXX**

Date:

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I **XXXXXXXX**, a Ph.D. student of the Academy of Scientific and Innovative Research (AcSIR) with Registration No. **YYYYYYYY** hereby undertake that, the thesis entitled **"Long Title of Thesis"** has been prepared by me and that the document reports original work carried out by me and is free of any plagiarism in compliance with the UGC Regulations on *"Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions (2018)"* and the CSIR Guidelines for *"Ethics in Research and in Governance (2020)"*.

Signature of the student
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It is hereby certified that the work done by the student, under my supervision, is plagiarism-free in accordance with the UGC Regulations on *"Promotion of Academic Integrity and Prevention of Plagiarism in Higher Educational Institutions (2018)"* and the CSIR Guidelines for *"Ethics in Research and in Governance (2020)"*.

Signature of the Supervisor
Name:
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Dedication

To those who inspired this journey and will never read this.

"When life gives you lemons, construct a crude electrochemical battery" - Unknown

Acknowledgement

Acknowledgement here

Your Name

Abstract in your mother tongue

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Abbreviations

AC	Alternating Current
AFM	Atomic Force Microscopy
DC	Direct Current
FESEM	Field Emission Scanning Electron Microscope
GITT	Galvanistatic Intermittent Titration Technique
HSE	Hybrid Solid Electrolyte
IL	Ionic Liquid
LAGP	Lithium Aluminium Germanium Phosphate, $Li_{1+x}Al_xGe_{2-x}(PO_4)_3$
MS	Mass Spectroscopy
NASICON	Sodium (Na) Super Ion Conductor
OCP	Open Circuit Potential
PAN	Polyacrylonitrile
QSE	Quasi-Solid Electrolyte
RTIL	Room Temperature Ionic Liquid
SCE	Solid Composite Electrolyte
TPU	Thermoplastic Polyurethane
UTM	Universal Testing Machine
XRD	X-Ray Diffraction

Chapter 1
Introduction

Write main matter of chapter here [1]

Figure 1.1

References

[1] Evan Kurian, Jayashree Pitchai, Soundarya Neelanarayanan, and K Ramesha. In pursuit of all solid state batteries (assb): advances at the cathode–electrolyte interface for garnet-based assb. *RSC Applied Interfaces*, 1(5):868–895, 2024.

Chapter 2
Methodology

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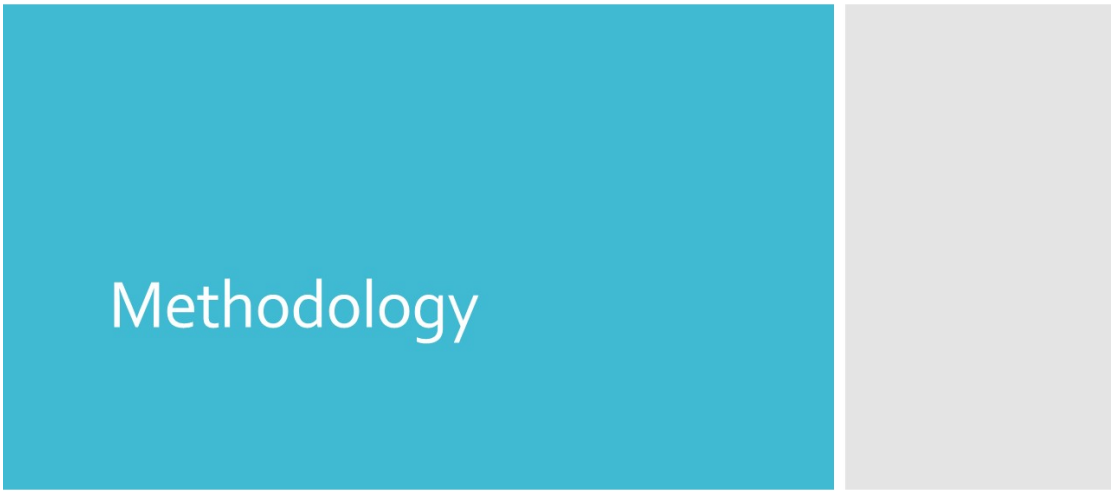

Figure 2.1

References

[1] Evan Kurian, Jayashree Pitchai, Soundarya Neelanarayanan, and K Ramesha. In pursuit of all solid state batteries (assb): advances at the cathode–electrolyte interface for garnet-based assb. *RSC Applied Interfaces*, 1(5):868–895, 2024.

Chapter 3
Short Title of Chapter 1

Long Title of Chapter 1[†]

Abstract

Write abstract here

[†]This work has been peer reviewed and published: ***Ionics***, 2022, 28(**6**):2649-2660

Write main matter of chapter here [1]

Figure 3.1

References

[1] Ramakumar Sampathkumar, MG Karthick Babu, Evan Kurian, RN Ramesha, Dasari Bosubabu, and K Ramesha. Influence of lithium metal anode coated with a composite quasi-solid electrolyte on stabilizing the interface of all-solid-state battery. *Ionics*, 28(6):2649–2660, 2022.

Chapter 4
Short Title of Chapter 2

Long Title of Chapter 2[†]

Abstract

Write abstract here

[†]This work has been peer reviewed and published: ***Journal of Energy Storage***, 2025, **115**:115882

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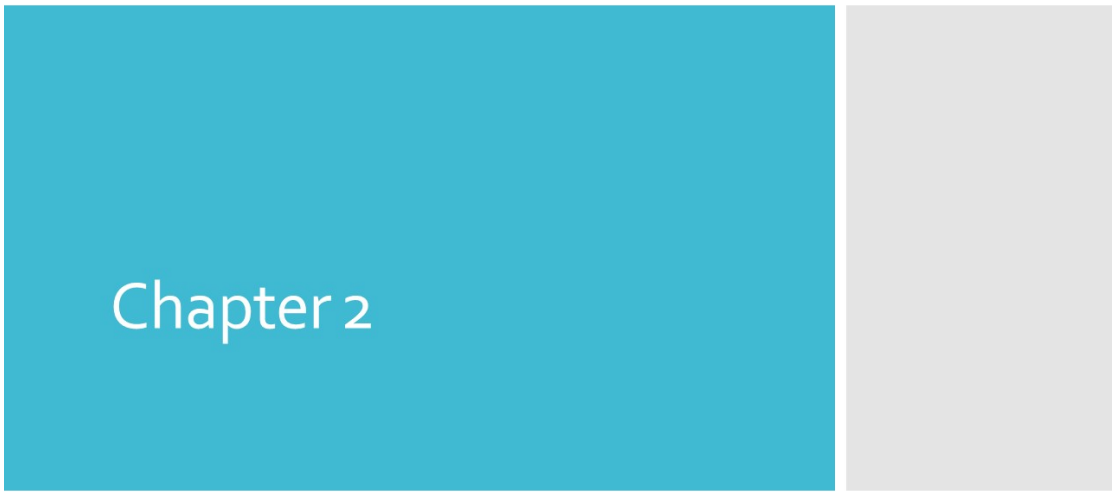

Figure 4.1

References

[1] Evan Kurian, Jayashree Pitchai, Soundarya Neelanarayanan, Deepak Kumar, Sellamuthu N Jaisankar, and K Ramesha. Thermoplastic polyurethane (tpu) based high-performing solid polymer electrolytes for solid-state lithium metal batteries. *Journal of Energy Storage*, 115:115882, 2025.

Chapter 5
Short Title of Chapter 3

Long Title of Chapter 3[†]

Abstract

Write abstract here

[†]This work has been peer reviewed and published: ***Physica B: Condensed Matter***, 2021, **615**:413087

Write main matter of chapter here [1]

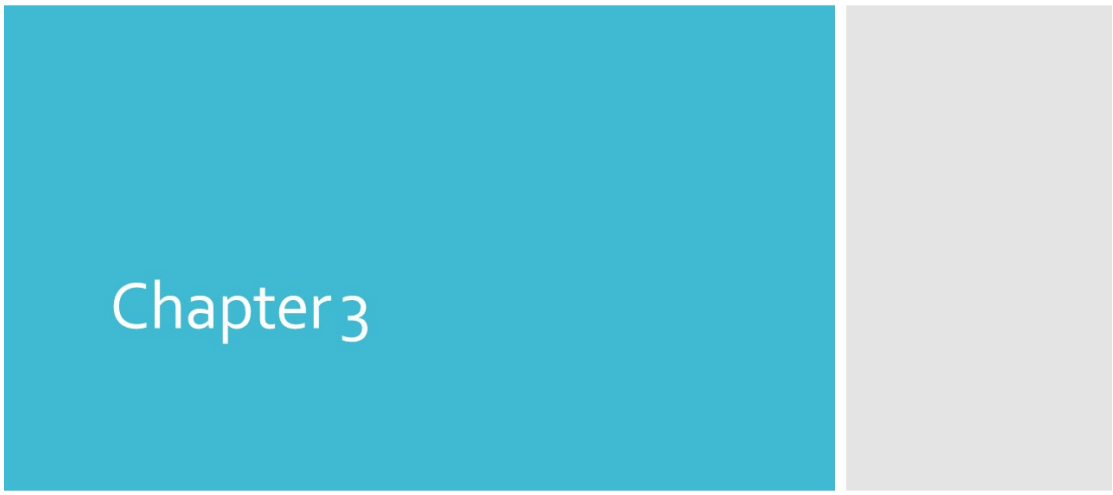

Figure 5.1

References

[1] P Muhammed Shafi, Evan Kurian, Nikhitha Joseph, Selvakumar Sellaiyan, Akira Uedono, and A Chandra Bose. Effect of ag doping on crystallinity and microstrain of lamno3 nanoparticles: confirmations of defect levels with positron lifetime and doppler-broadening calculations. *Physica B: Condensed Matter*, 615:413087, 2021.

Chapter 6
Conclusion

Main matter of conclusion here

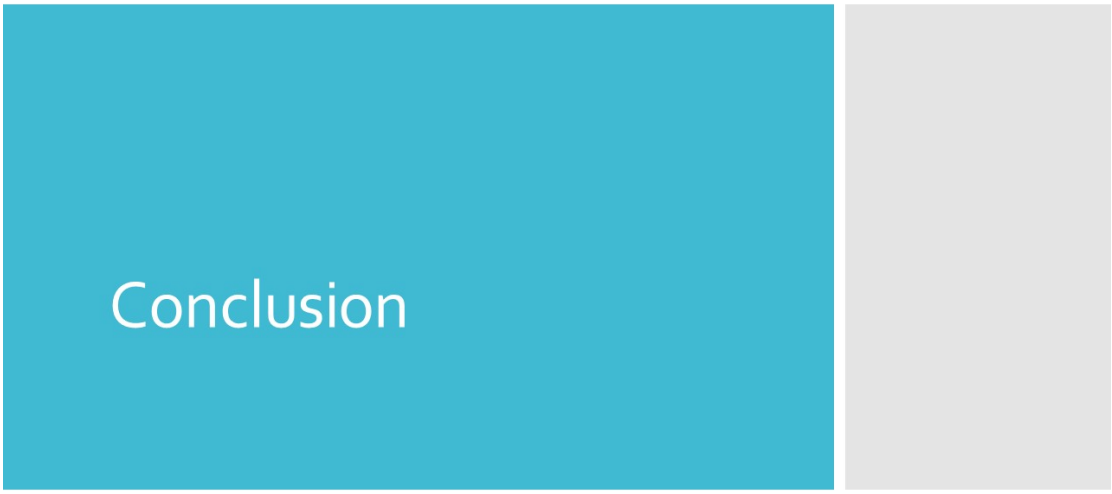

Figure 6.1

Abstract

Name of the Student: **XXXX XXXXXX** Registration No.: **YYYYYYYYYYYY**
Faculty of Study: **Physical Science** Year of Submission: **2025**
AcSIR academic center / CSIR Lab: **ZZZZZZZZZZ** Name of the Supervisor: **Dr. XXXXXX**
Title of the thesis: **Long Title of Thesis**

Thesis abstract here

Appendix I

List of Publications in SCI Journals emanating from the thesis work

1. **Evan Kurian**, Jayashree Pitchai, Soundarya Neelanarayanan, Deepak Kumar, Sellamuthu N. Jaisankar, and K. Ramesha. ”Thermoplastic polyurethane (TPU) based high-performing solid polymer electrolytes for solid-state lithium metal batteries.” *Journal of Energy Storage* 115 (2025): 115882

2. **Evan Kurian**, Jayashree Pitchai, Soundarya Neelanarayanan, and K. Ramesha. ”In Pursuits of All Solid State Batteries (ASSB): Advances at the Cathode-Electrolyte Interface for Garnet-Based ASSB.” *RSC Applied Interfaces* 1.5 (2024): 868-89

3. Sampathkumar, Ramakumar, M. G. Babu, **Evan Kurian**, R. N. Ramesha, Dasari Bosubabu, and K. Ramesha. ”Influence of lithium metal anode coated with a composite quasi-solid electrolyte on stabilizing the interface of all-solid-state battery.” *Ionics* (2022): 1-12.

4. **Evan Kurian**, Jayashree Pitchai, Deepak Kumar, Shreya Jose, and K. Ramesha. ”Plasticizer-Assisted Garnet LLZO Integration in TPU-based Solid Composite Electrolytes for High-Performance Solid-State Lithium Metal Batteries.” *Manuscript accepted*

Appendix II

List of Papers presented in National / International conferences

1. Presented paper

Title: Thermoplastic Polyurethane (TPU) based high-performing Solid Polymer Electrolytes for Solid-State Lithium Metal Batteries
Conference: 13th International Symposium on Advances in Electrochemical Science and Technology (iSAEST-13), January 8-10, 2025, Thiruvananthapuram, Kerala, India
Abstract:
Abstract here

2. Presented poster

Title: Influence of lithium metal anode coated with a composite quasi-solid electrolyte on stabilizing the interface of all-solid-state battery
Conference: International Conference on Energy Conversion and Storage (IECS-2023), 18-20 January 2023, Research Park, IIT Madras, Chennai, India
Abstract:
Abstract here